Flood Watch Punjab:
satellite flood detection and reservoir level forecasting

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Abstract

Effective and accurate flood detection and warning systems require timely inundation detection with a reliable and constantly available source of information on reservoir conditions. This study aims to assess floods in the region of Punjab and provide an accurate flood mapping using Sentinel-1 Synthetic Aperture Radar(SAR) observations of Punjab processed through a custom pipeline to refine classifications and also determine whether combining satellite prediction signals with river catchment derived signals improved inundation prediction. Independently evaluated fixed reduction of Copernicus Global Flood Monitoring(GFM) data for 52 locations using SAR and Landsat-9 observations of a total of 673.95 km2 of usable coverage out of a total of 1,577.16km2 of total coverage showed that, among the locations with conclusive image based references, the sampling-weighted precision and recall were 93.8% and 66.1%, respectively. Reservoir forecasts for Bhakra and Pong reservoirs were assessed chronologically through 2021-24 and tested against persistence and a matching undamped pattern. The river reservoir prediction showed for Bhakra, the MAE (mean absolute error) value was 0.247m meanwhile for Pong it was 0.916m. Bhakra improved over persistence for all 5 horizons showing a 36.6% MAE reduction over the 5th horizon. The encountered bottlenecks which led to difficulties and a higher MAE include but are not constrained by the limited access to real-time reservoir and satellite data. Early and accurate prediction along with a system of rapid response to predictions would save structures and prevent livestock, ,monetary and agricultural damages.

1 Introduction and research questions

Satellite observations can show inundated land, reservoir records describe stored water, and weather forecasts estimate future rainfall. Their combination is useful only if a resulting warning can be compared with a later flood observation. A reservoir-level forecast alone does not identify where land will flood: releases, river routing and local rainfall also matter. Punjab's agricultural plains and the Satluj-Beas system motivate preserving the spatial footprint and time of each source [6].

Satellite observation pipelines uses the existing GFM mapping service and its contextual and quality layers [1, 2, 3]. The evaluated component screens those classifications. The river reservoir prediction pipeline's separate historical benchmark uses a frozen damped observation(recent). The integration experiment compares input signals through logistic regression.

The questions are whether fixed GFM classifications agree with near-time reviewed inundation; whether a chronological level forecast improves on persistence and what damping adds; whether river-informed inputs improve later-reference warning decisions over satellite history; and which spatial, temporal and reference gaps restrict broader claims.

1.1 Monetary losses and effects over the following year

The August–September 2025 floods in Punjab, India damaged crops, housing and public infrastructure. Punjab's relief portal initially estimated losses at Rs 13,800 crore [18]. A 10 January 2026 report on the revised state assessment gives Rs 11,855.65 crore (about Rs 118.56 billion) in reconstruction demand, including Rs 3,949 crore for roads, Rs 3,034 crore for agriculture and horticulture, and Rs 2,077.61 crore for water resources and irrigation [19]. These are reported assessment figures, not losses calculated by Flood Watch or a measured one-year GDP reduction.

Over the following year, lost harvest income and the cost of replacing seed, livestock and equipment can constrain household recovery. Residual silt, waterlogging and damaged irrigation can disrupt the next sowing cycle; a September 2025 government field visit documented both destroyed crops and concern for the next crop [20]. Road and market repairs, housing recovery and restoration of schools and health facilities can extend the burden beyond the flood season [18]. These are plausible recovery pathways, not a completed twelve-month impact survey. The project has not measured monetary losses avoided by its warnings.

2 Study area and Data

The project covers Punjab, Bhakra and Pong catchments, and selected downstream reaches. Catchment aggregation uses retained HydroBASINS-derived geometry [4]. Satellite summaries use twenty commit-pinned Census-2011 project polygons [5]; these historical units do not reproduce all twenty-three current administrative districts [6]. Missing current district names do not indicate observed dryness.

The detection case uses Sentinel-1 sensing near 01:00 UTC and Landsat-9 at 05:37 UTC on 16 September, with earlier and bracketing Sentinel-2 context. The geometric pilot is 1,577.16 km^2; 673.9488 km^2 has joint usable support at the 60 m comparison grid. That support is comparison coverage, not flooded area. The retained CWC-derived archive contains monsoon-period records dated 2015–2025 [7]. Unit, finite-value, range and conflict checks accept 1,064 dated Bhakra and 830 Pong levels in native metres (Figure 1).

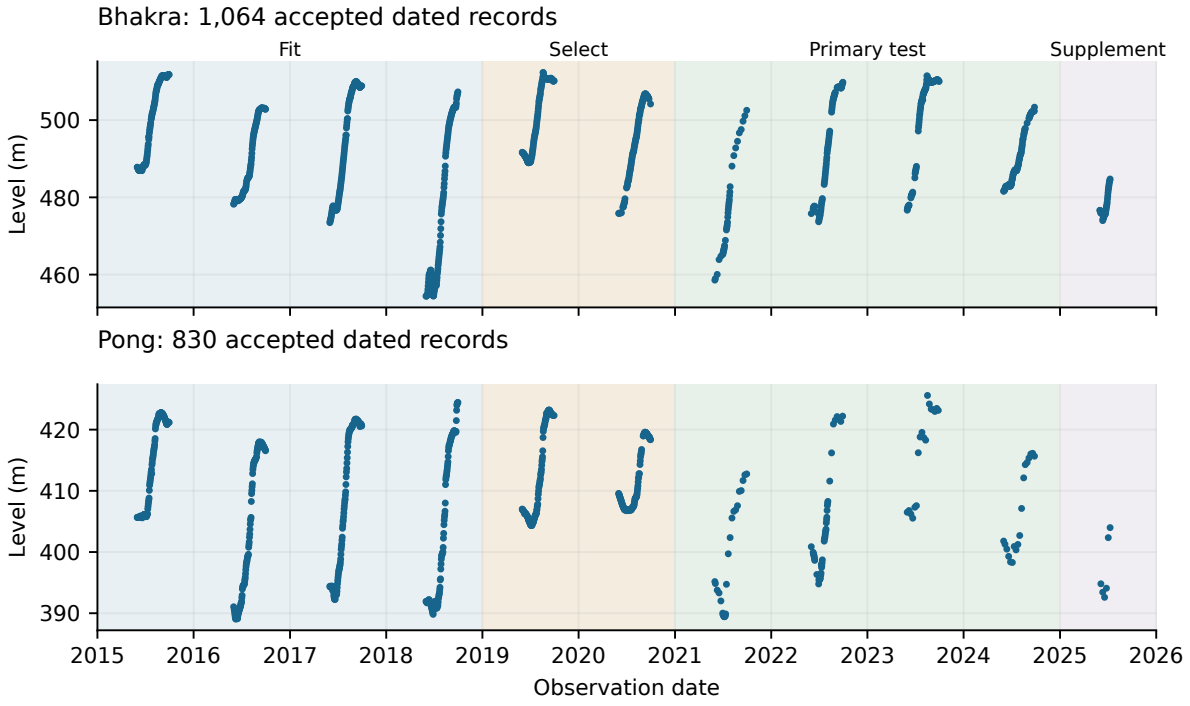


Figure 1: Accepted dated reservoir levels: Bhakra $n = 1,064$, Pong $n = 830$. Gaps are not interpolated. Shading distinguishes fitting, selection, primary testing and the incomplete 2025 supplement.

Table 1: Source roles. Source access, modeled history and reference interpretation are distinct from measured predictive accuracy.

Source	Retained quantity	Evaluation role
Sentinel-1/GFM	Flood, water, likelihood and contextual rasters	Fixed reduction and acquisition-specific classified area
Landsat-9/Sentinel-2	Quality-screened reflectance and dated image context	Reviewed imagery and separate new-water proxies
CWC archive / BBMB	Dated metres / printed feet and cusecs	Historical level targets / pre-issue reservoir reports
NRSC / NOAA GFS	Mapped inundation / original pre-issued rain forecasts	Later map target / 24- and 72-hour catchment inputs
IFS/AIFS / model history	Fixed weather runs / stitched historical forecasts	Prospective daily-inflow scenarios; history is modeled

2.1 Geographic coverage across Punjab

Ten of twenty historical project polygons have usable classifications (Figures 2–3). Latest usable shares range from 1.70% to 89.78%, with median 22.91%. Nine selected summaries are from 4 October 2026 and Firozpur from 5 October; they are not simultaneous observations. Historical boundaries include later district splits and differ from the twenty-three current districts. All usable flood fractions are below 0.5%, but unknown coverage remains unresolved: Firozpur's selected acquisition classifies only 5.18% of its polygon.

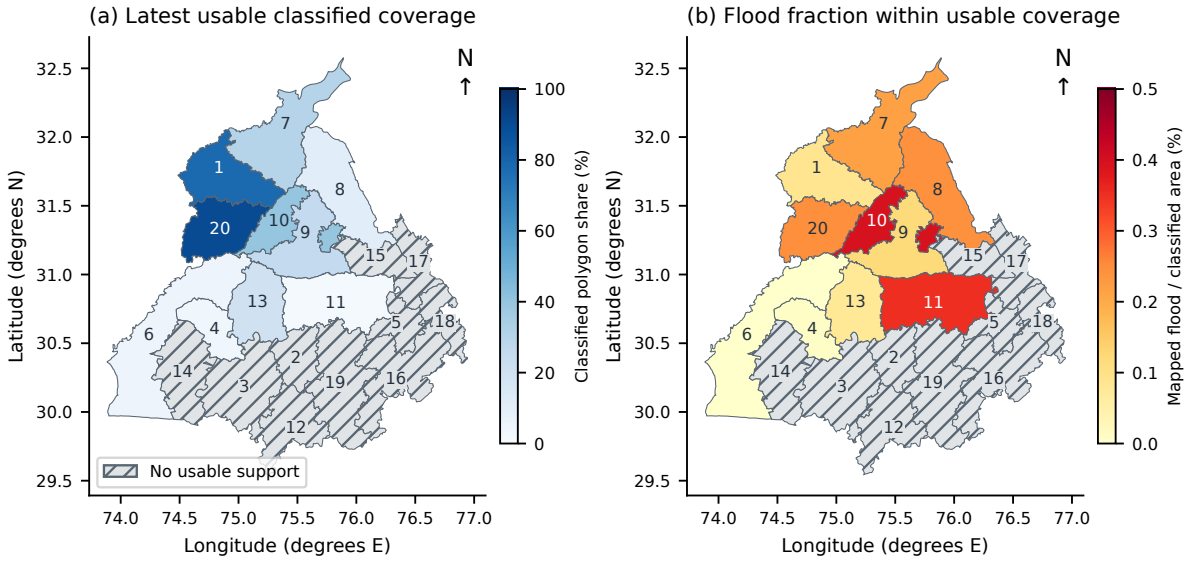


Figure 2: Punjab choropleths: (a) classified polygon share and (b) mapped flood within usable support, using latest usable acquisitions on 4–5 October 2026. Colors are polygon summaries, not floodwater locations or depths. Hatching means unknown support; numbers match Figure 3.

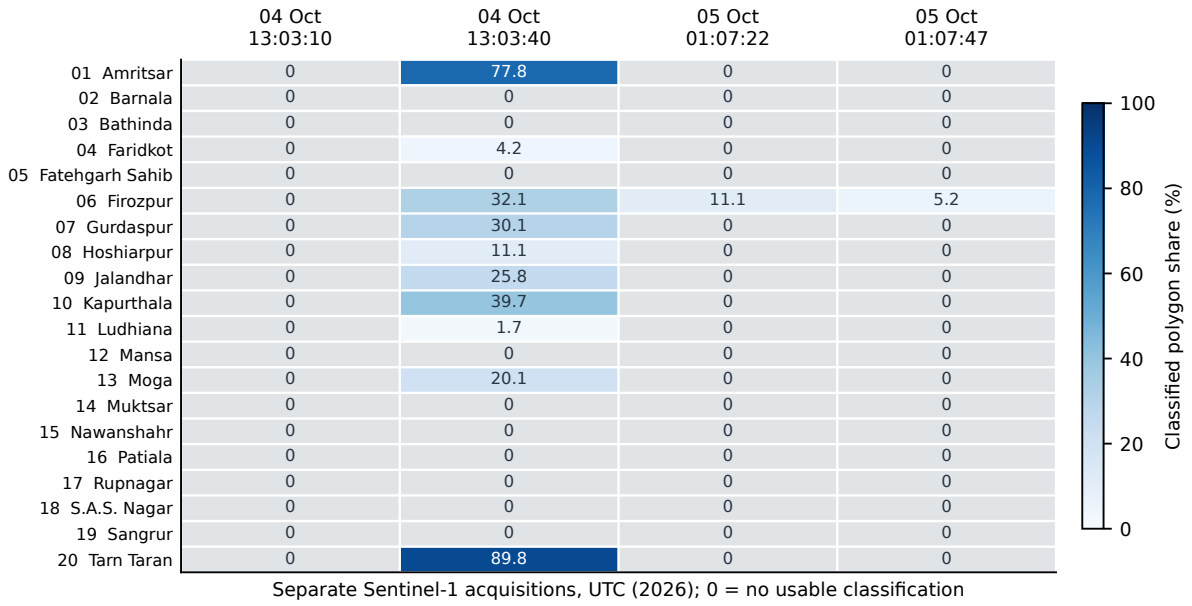


Figure 3: Coverage heat map by historical polygon and acquisition time (UTC): classified polygon percentages, with twelve nonzero cells out of eighty. Grey zeros mean unavailable classification, not dry land. Acquisitions are kept separate.

3 Methods

3.1 Fixed satellite reduction and spatial support

The local detector requires six acquisition-linked layers: i)ensemble ii)flood extent, iii)water extent, iv)exclusion, v)reference water, vi)likelihood and advisory flags. Missing, excluded, advisory and reference-water pixels remain unknown. A retained pixel must agree between flood class and likelihood relative to 50; a flood-positive pixel must also have water extent one. These were taken as local rules for consistency. A provider anomaly flag is significance metadata rather than an additional quality failure. Reference water withholding is retained because the captured metadata do not independently establish the selected monthly reference mask.

For unit d and acquisition a, let a_p be pixel area intersecting its polygon A_d, v_p indicate usable support and f_p identify mapped flood. Classified area V, flood area F, classified share c and within-support flood fraction q are

$$V_{d,a} = \sum_p a_p v_p, \qquad F_{d,a} = \sum_p a_p v_p f_p, \tag{1}$$

$$c_{d,a} = V_{d,a}/|A_d|, \qquad q_{d,a} = F_{d,a}/V_{d,a} \quad (V_{d,a} > 0). \tag{2}$$

Acquisitions are not pooled. The exploratory classified-support event rule is $y_{d,a} = \mathbf{1}\{q_{d,a} > 0.005\}$, with no additional minimum-coverage cutoff. Thus the associated c must accompany a label. With unknown area $U = |A_d| - V$, a compatible whole-polygon flood fraction π satisfies

$$L = F/|A_d| = cq \le \pi \le H_u = (F+U)/|A_d| = cq + 1 - c. \tag{3}$$

These are conditional logical bounds, not confidence intervals. Unknown pixels cannot be counted as dry land.

3.2 Reference review, sampling and agreement scores

The 4.62-hour radar/optical pairing supports near-time comparison but permits change between observations. Optical cloud, shadow, saturation, aerosol and invalid-reflectance exclusions follow retained quality contracts and Landsat QA interpretation [12]. A common 60 m cell requires consistent native support; mixed cells are withheld. Water indices generate candidates, not automatic flood truth. Earlier and bracketing imagery distinguish ordinary water, agricultural wetness and possible land inundation.

One eligible cell is sampled per selected 600 m stratum/block, with seed 20250916. All thirteen positive blocks and nineteen nearby-negative blocks, plus twenty of 2,913 background blocks, are sampled (Table 2). The recorded inclusion probability p_i accounts for block and within-block selection; design weight is $w_i = 1/p_i$.

Table 2: September sampling frame at 60 m: 187,208 eligible cells give 673.9488 km^2 of joint comparison support.

Stratum	Eligible cells	Eligible blocks	Sampled blocks
Detector positive	88	13	13
Nearby negative	229	19	19
Background negative	186,891	2,913	20

For decided references, confusion counts T_P, F_P, F_N, T_N give

$$P = \frac{T_P}{T_P + F_P}, \qquad R = \frac{T_P}{T_P + F_N}, \tag{4}$$

$$F_1 = \frac{2T_P}{2T_P + F_P + F_N}, \qquad \mathrm{IoU} = \frac{T_P}{T_P + F_P + F_N}. \tag{5}$$

Weighted ratios replace counts by sums of w_i in each category. A zero denominator is unavailable. Ordinary water is saved separately and counts as non-flood only in the September binary diagnostic; uncertain references are excluded. Weighted totals are sampling estimates, not additional observations or measured flood area. One sample per stratum/block does not provide design-unbiased within-block variance, and correlated locations do not justify naive binomial intervals. The user's review had AI proposals visible; it supplies interpreted-image agreement, not blinded expert or field truth.

3.3 Chronological reservoir-level experiment

Fitting ends on 31 December 2018; model selection uses 2019–2020; primary testing uses 2021–2024. This was done to make sure we have enough data for training. Source dates register candidate issues. At issue-date label t, only levels through $t-1$ are allowed, and eight exact consecutive daily records are required. Missing histories or targets are withheld without interpolation. Source checks require native metres, finite values and no same-date conflict. Broad range gates are 412.06–522.06 m for Bhakra and 323.67–433.67 m for Pong.

For lookback $k \in \{1, 3, 7\}$ and nominal horizon $H \in \{1, \ldots, 5\}$,

$$g_{t,k} = \frac{z_{t-1} - z_{t-1-k}}{k}, \qquad \widehat{z}_{t+H} = z_{t-1} + \alpha_{k,H}(H+1)g_{t,k}. \tag{6}$$

Persistence holds z_{t-1}. For fitting rows, set $x_i = (H+1)g_{i,k}$ and $u_i = z_{i+H} - z_{i-1}$. The constrained least-squares damping coefficient is

$$\widehat{\alpha}_{k,H} = \mathrm{clip}_{[0,1]}\left(\frac{\sum_i x_i u_i}{\sum_i x_i^2}\right), \tag{7}$$

with zero damping if the denominator is zero. Persistence and three damped candidates are compared on identical selection cases. Minimum selection MAE freezes a separate choice per dam/horizon; ties prefer persistence, then shorter lookback. Table 3 records the resulting coefficients and counts. The source audit reconstructs all forty candidate fits and selection scores within 10^{-12}.

Table 3: Frozen selected models. Errors are metres; k is days. Fit and selection periods precede primary testing.

Dam	H	k	α	Fit n	Select n	Select MAE
Bhakra	1	1	0.8233	401	176	0.206
Bhakra	2	3	0.8505	397	175	0.322
Bhakra	3	3	0.8259	393	174	0.438
Bhakra	4	3	0.8045	389	172	0.557
Bhakra	5	3	0.7818	385	172	0.690
Pong	1	1	0.6469	389	199	0.187
Pong	2	3	0.7213	385	196	0.287
Pong	3	3	0.7235	381	195	0.388
Pong	4	3	0.7186	377	193	0.486
Pong	5	3	0.7075	373	191	0.588

The last known record is dated $t-1$, so $H = 1$ spans two known-to-target calendar-date labels. Actual observation-hour leads and historical publication availability are unknown. The one-date buffer is an experimental assumption. A post-hoc undamped ablation retains the selected k and sets $\alpha = 1$; it is not a separately tuned competitor.

For paired error $e_i = \widehat{z}_i - z_i$ and persistence error e_i^{pers},

$$\mathrm{MAE} = n^{-1}\sum_i |e_i|, \qquad \mathrm{RMSE} = \sqrt{n^{-1}\sum_i e_i^2}, \qquad \mathrm{Bias} = n^{-1}\sum_i e_i, \tag{8}$$

$$\Delta_{\mathrm{MAE}} = n^{-1}\sum_i (|e_i| - |e_i^{\mathrm{pers}}|). \tag{9}$$

Negative Δ favors the damped trend. Whole test-year blocks are resampled 10,000 times, using the same case-weighted statistic $\Delta^* = \sum_{y\in B} L_y / \sum_{y\in B} n_y$. Central 95% percentile intervals are exploratory with four Bhakra blocks; fewer than three blocks receive no interval. Common-cohort and omitted-year checks retain frozen coefficients and are post-hoc.

3.4 Historical integration design and feature comparison

The target is flood status at a later reference snapshot, approximately 53.6–60.5 hours after nominal issue (Table 4). It does not prove continuous flooding or dryness throughout three days.

Table 4: Integration issue and reference clocks, UTC. H23A/H23B are training; H24/H25 supply later testing.

Snapshot	Issue	Reference target	Lead (h)
H23A	2023-08-13 00:00:00	2023-08-15 05:36:49.024000	53.614
H23B	2023-08-18 00:00:00	2023-08-20 05:36:51.024000	53.614
H24	2024-08-22 00:00:00	2024-08-24 05:36:41.024000	53.611
H25	2025-08-25 00:00:00	2025-08-27 12:30:00	60.500

The nineteen eligible training cases are from 2023: eleven flood and eight non-flood. The twenty common later cases comprise four 2024 optical non-water controls and sixteen 2025 NRSC map targets, ten mapped positive and six map absence. Two reviewed cases lack usable past satellite state and remain excluded. The NRSC map refers to Sentinel-1A sensing on 27 August at 18:00 IST and was issued the next day [14]. Its red areas are mapped inundation; white areas are no mapped inundation, not observed physical dry land.

All three arms include two local coordinates. Satellite history adds last known class, observation age, neighbourhood flood fraction and neighbourhood known fraction. River-informed inputs add 24- and 72-hour NOAA GFS rainfall over five linked catchments and level, reported outflow and measurement age for Bhakra and Pong [13]. Rainfall is mm; reports retain feet, cusecs and days. Original model initialization, forecast intervals and object publication clocks precede issue cutoffs. Unspecified flow averaging periods are not converted to volumes.

Table 5: Primary input arms. Counts are retained columns; shared coordinates appear in every arm.

Arm	Columns	Additional input groups
Satellite history	6	Last class, its age and two neighbourhood fractions
River-informed	18	Ten catchment rain forecasts and six reservoir report fields
Combined	22	All satellite-history and river-informed columns

Training-only median imputation and standardization produce $\widetilde{x}$. Balanced L2 logistic regression, scikit-learn 1.9.1, uses $C = 0.1$, lbfgs, maximum 2,000 iterations and random state 0 [17]:

$$\widehat{p}_i = \frac{1}{1 + \exp[-(\beta_0 + \beta^\top \widetilde{x}_i)]}, \qquad \widehat{y}_i = \mathbf{1}\{\widehat{p}_i \geq 0.5\}. \tag{10}$$

The 0.5 cutoff is predicted probability, distinct from the 0.5% spatial-fraction rule. No test-driven threshold or hyperparameter search is used. These enriched samples do not establish probability calibration. Current all-season hydrological parameters are excluded; this compares selected signals rather than replaying the full inflow/storage/release/routing engine. Only one nonmissing reservoir-report state occurs in training, so its separate contribution is not identifiable.

Forty-one references received a second user review with suggestions and earlier labels hidden, shuffled IDs and required reasons. Because initial results were already disclosed, this improves review procedure without restoring an unseen test. Original-label training, audited training and exclusion of the unresolved ordinary-water case R006 are retained sensitivity fits.

4 Results

4.1 September imagery agreement

The user reviewed all fifty-two sampled locations: ten flood, thirty-eight non-water, one ordinary water and three uncertain. Among forty-nine decided references, the detector identifies eight floods, misses two, flags two non-flood cells and leaves thirty-seven non-flood cells unflagged. All three uncertain references are detector positives and remain outside decided-only scores.

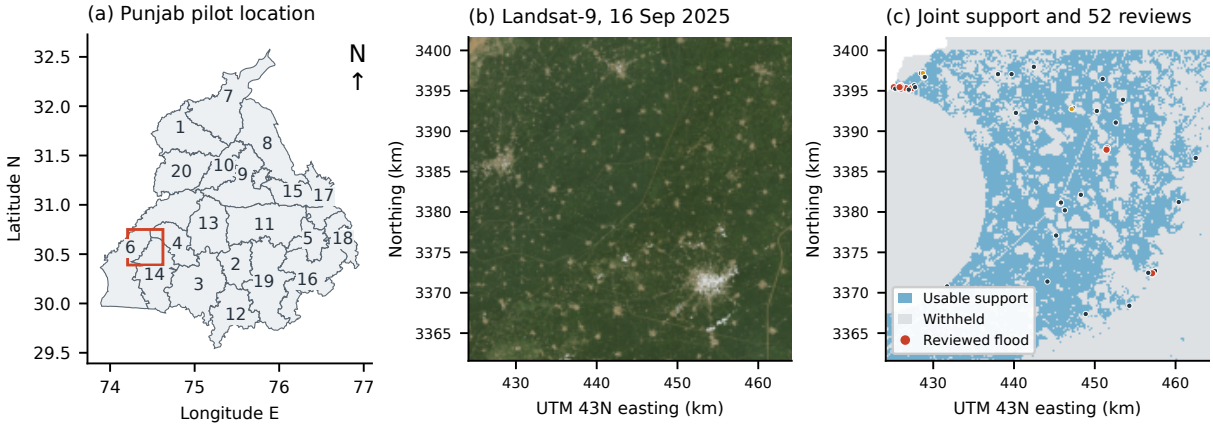


Figure 4: September pilot: (a) location, (b) Landsat-9 true color, and (c) usable 60 m support with 52 reviews. Blue support totals 673.95 km^2 in the 1,577.16 km^2 pilot; grey areas are withheld. Points denote flood (red), non-water (dark), ordinary water (purple) and uncertain (yellow) references, not flood extent.

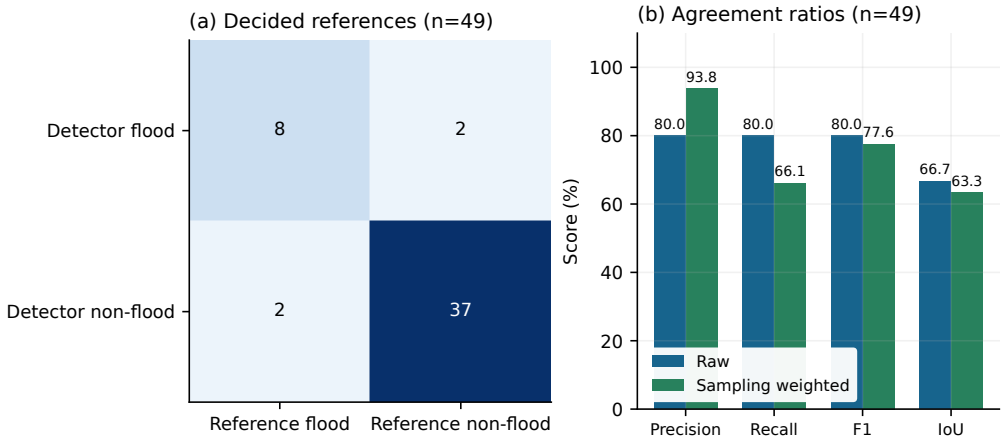


Figure 5: September comparison, $n = 49$ decided references: complete confusion counts and raw versus sampling-weighted agreement. The three uncertain locations are unscored; the weight totals are sampling quantities, not additional cases.

Weighted confusion totals are TP 76, FP 5, FN 39 and TN 149,918.2, plus uncertain weight seven. The missed references carry weights four and thirty-five, reducing weighted recall relative to raw recall. Treating uncertain labels as all non-flood or all flood gives weighted precision 86.4–94.3% and recall 66.1–68.0%; these are label-sensitivity ranges, not confidence intervals. Overall accuracy is dominated by dry-background weight and is not the headline measure.

All eight reviewed matches are S001, S012, S016, S026, S034, S037, S038 and S050. Their coordinates remain in the companion ledger. Misses are S004/S005; false alarms are S007 (non-water) and S043 (ordinary water). S035/S042/S046 remain uncertain. Several adjacent matches can belong to one inundation feature, so eight cells are not eight independent events. The detector, sampled locations and original thresholds were not retuned to improve these matches.

4.2 Primary reservoir-level errors

Bhakra's frozen damped trend has lower MAE than persistence at every primary horizon. At $H = 1$, MAE falls from 0.488 to 0.247 m on 194 paired cases, a 49.3% reduction. At $H = 5$, 0.761 versus 1.201 m gives a 36.6% reduction on 167 cases. The undamped first-horizon result is 0.259 m: damping adds only 0.012 m, approximately 4.5%, beyond the recent trend. Most of the gain over persistence is therefore attributable to trend extrapolation.

Table 6: Primary dated-level errors, 2021–2024. All errors are metres. All comparators use the same paired cases within a row; H implies $H + 1$ known-to-target calendar days.

Dam	H	n	Damped MAE	Pers. MAE	Undamped MAE	RMSE	Bias
Bhakra	1	194	0.247	0.488	0.259	0.423	-0.075
Bhakra	2	189	0.381	0.701	0.400	0.577	-0.080
Bhakra	3	181	0.509	0.886	0.549	0.753	-0.110
Bhakra	4	174	0.644	1.054	0.722	0.932	-0.142
Bhakra	5	167	0.761	1.201	0.878	1.096	-0.175
Pong	1	12	0.916	1.068	1.122	1.083	-0.364
Pong	2	11	0.785	1.457	0.646	0.897	-0.594
Pong	3	9	1.052	1.887	0.800	1.160	-0.827
Pong	4	6	1.409	2.647	0.924	1.573	-1.409
Pong	5	4	1.838	3.090	1.320	1.980	-1.838

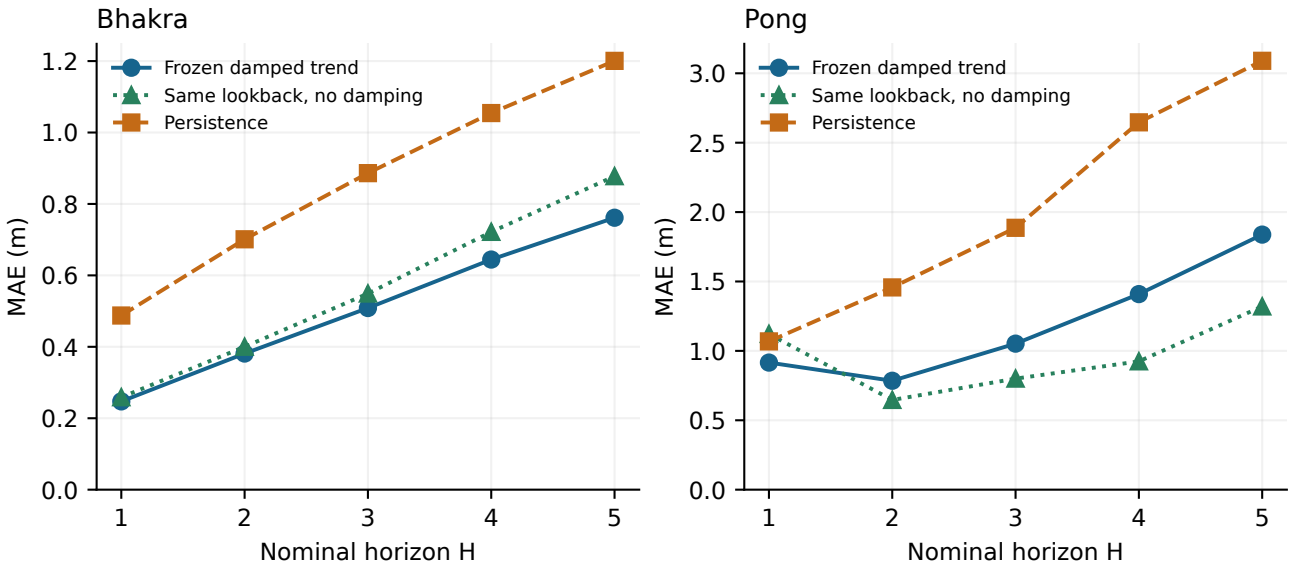


Figure 6: Frozen damped trend, persistence and selected-lookback undamped ablation. Bhakra paired n at $H = 1$–5: 194/189/181/174/167; Pong: 12/11/9/6/4. Lines connect aggregate errors, not additional observations.

Pong's first-horizon MAE is 0.915708 m, rounded to 0.916 m, across twelve cases. Its undamped comparator has lower MAE than the frozen damped model at horizons 2–5. Damping therefore does not transfer consistently between reservoirs. Pong's decreasing sample counts, down to four at $H = 5$, cannot establish stable future performance. No coefficients are retuned after these adverse comparisons.

Across two dams and five horizons, 947 paired rows survive from 4,880 registered primary rows: 3,745 lack complete history and 188 lack a target. The paired rows occupy 197 distinct issue-date labels, but dates, issues and horizons overlap. These are not 947 independent events. Apparent horizon-to-horizon change also mixes different eligible cases; the controlled comparison is model versus comparator within a horizon. Historical level errors do not establish inflow, discharge or downstream inundation accuracy.

4.3 Year effects, robustness

Bhakra $H = 1$ improves over persistence in each primary year, but annual gains differ sharply (Figure 7). The six-case 2021 year improves by about 1.196 m; the sixty-nine-case 2023 year improves by only 0.079 m. Case weighting keeps the short year from dominating the mean. Model median absolute error is 0.131 m versus persistence's 0.320 m; empirical 90th percentiles are 0.606 versus 1.177 m. These distributions are descriptive because daily cases overlap.

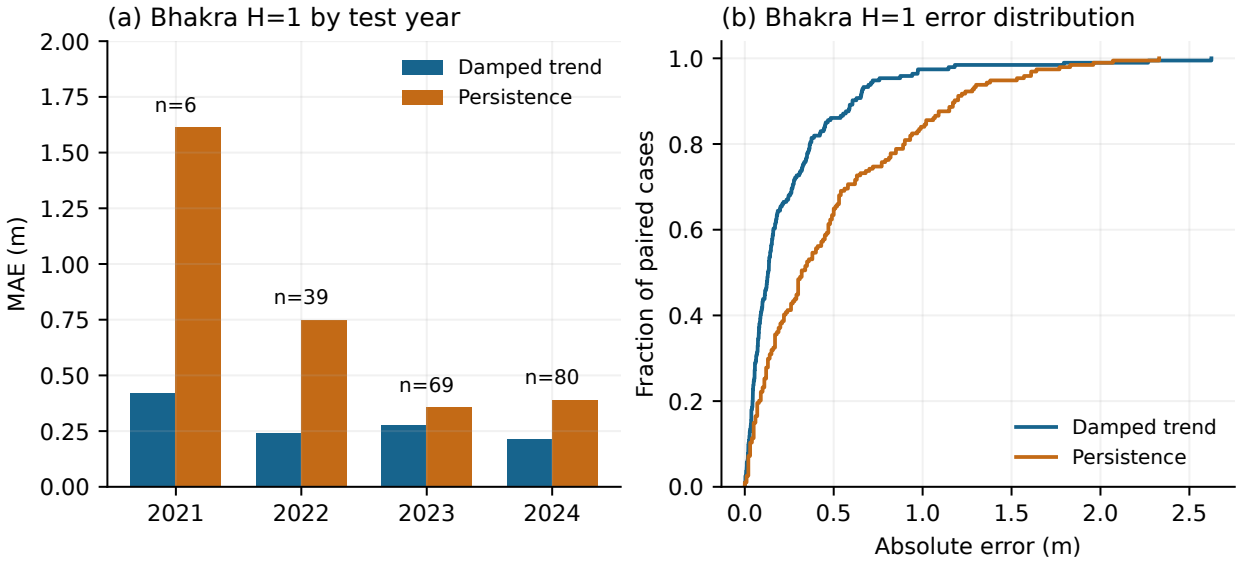


Figure 7: Bhakra $H = 1$ annual MAE and empirical error distribution, total $n = 194$. Annual n: 6, 39, 69 and 80. Paired dated errors do not constitute independent events.

The largest model absolute error, 2.624 m, exceeds persistence's maximum of 2.330 m. For the target dated 1 August 2023, the model predicts 504.126 m against 506.750 m, while persistence predicts 505.180 m. The recent trend fails to anticipate this increase. Lower average error does not imply better prediction during every rapid change.

Bhakra's first-horizon bias is −0.075 m. Its paired MAE difference is −0.240 m, with corrected exploratory year-block interval $[-0.601, -0.111]$ m. Four blocks leave substantial transfer and provenance uncertainty. Pong has fewer than three contributing blocks at every horizon and receives no comparable interval.

Table 7: Corrected Bhakra paired loss differences and exploratory 95% whole-year bootstrap intervals. Units are metres; four blocks, 10,000 draws. Negative differences favor the damped model.

H	Paired n	Difference	Lower 95%	Upper 95%
1	194	-0.240	-0.601	-0.111
2	189	-0.320	-0.842	-0.128
3	181	-0.378	-1.117	-0.135
4	174	-0.410	-1.371	-0.120
5	167	-0.439	-1.610	-0.126

On identical 161 Bhakra issue dates across all horizons, model MAE rises from 0.249 to 0.774 m and persistence from 0.430 to 1.180 m. Omitting each primary year leaves $H = 1$ paired differences between −0.330 and −0.173 m without refitting; omitting 2021 gives −0.210 m on 188 cases. These post-hoc checks reuse the archive and condition on target availability.

4.4 Integration does not improve warning decisions

At the unchanged probability threshold of 0.5, none of the three primary arms flags any of the ten NRSC mapped-positive test cases. All produce six correct map-absence nonalerts and four optical-control nonalerts (Table 8). Precision is undefined because there are no positive predictions; NRSC mapped-positive recall is zero. Optical flood recall is undefined because the four optical test controls contain no flood reference. Source types are not pooled into physical flood accuracy.

Table 8: Final audited integration, twenty later cases per arm. NRSC white-map space is map absence, not physical dry land.

Input arm	Map TP	Map FP	Map FN	Map TN	Optical FP	Optical TN
Satellite history	0	0	10	6	0	4
River-informed	0	0	10	6	0	4
Combined	0	0	10	6	0	4

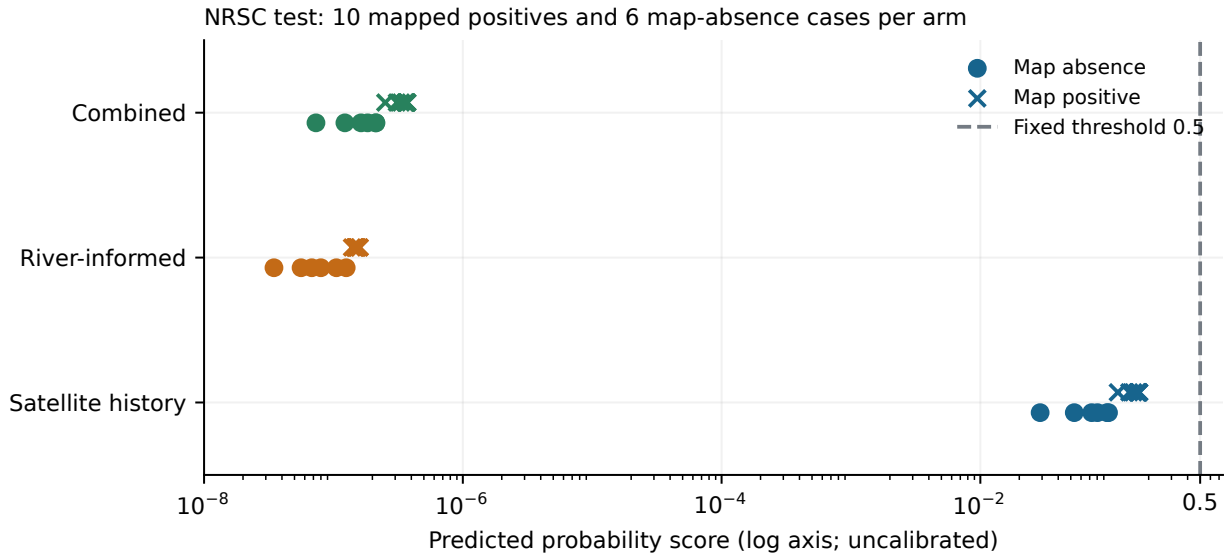


Figure 8: Audited fit: all sixteen NRSC test scores per arm on a logarithmic axis. Crosses: ten mapped positives; circles: six map-absence cases. Every probability is below 0.5. Scores are uncalibrated; overlapping markers may represent equal values.

The second review preserves twelve training-water candidates as flood and all ten NRSC positive choices as map labels. Twenty-three of twenty-five optical classes exactly match the first review; no previously decided optical binary class changes. R031 is corrected to vegetation/non-water, and R006 remains ordinary water with unresolved origin. Original-label training (nineteen rows), audited training (nineteen) and R006 exclusion (eighteen) all retain the same adverse warning decisions. Independent reconstruction of all 180 scores across these fits agrees with the archive within 9.8×10^{-17}.

The selected map-positive scores rank above map-absence scores, producing selected-case ROC AUC and average precision of 1.0 for each arm. This is compatible with zero recall at 0.5: ranking and threshold decisions answer different questions. Shared coordinates, mapped-positive supplementation and a tiny cohort prevent interpreting that perfect ranking as transferable warning performance. Lowering the threshold after seeing the test would be outcome-driven tuning and is not used.

In the preserved initial fifteen-case comparison, geography alone identifies ten initially positive references but flags two of five others. Weather-only, satellite-history, river-informed, combined, no-press-report and past-state persistence controls all miss the ten positives. That original set differs from the final source-separated twenty-case audit; it is retained as a selected-case diagnostic. Geography can discriminate an enriched sample without learning future flood dynamics.

5 Discussion and limitations

The September study records eight reviewed flood detections, two misses and two false alarms. Unequal sampling raises weighted precision to 93.8% and lowers recall to 66.1%, because missed references represent more eligible cells. The 4.62-hour pairing improves temporal relevance but permits real change.

The NRSC thematic map is external to the predictor, yet is itself satellite interpretation and may share Sentinel-1 error mechanisms with GFM. Map absence is not observed dryness. The second label audit followed disclosure of initial results and cannot restore an unseen evaluation. Positive enrichment, common coordinates and one training episode leave substantial spatial and source confounding. Probabilities are uncalibrated, and effectively constant training reservoir fields cannot isolate the reservoir contribution. Perfect ranking of selected cases alongside zero fixed-threshold recall reinforces the distinction between discrimination and useful warnings.

The Bhakra level result is a separate measured component gain. Recent trend accounts for most of its improvement over persistence; damping adds less and is inferior to the matched undamped comparator for Pong at horizons 2–5. Sparse target availability, particularly at Pong, can systematically exclude outages or rapidly changing periods. The historical source lacks observation hours, publication clocks and independently verified datum. A one-date buffer discloses an assumption rather than proving operational availability. Four-year bootstrap intervals exclude these provenance uncertainties; overlapping horizons cannot increase independent replication.

A further evaluation should freeze predictors before new outcomes, recruit independent reference interpreters, retain uncertain and ordinary-water classes, and cover multiple flood and non-event periods. It should pair warnings with later georeferenced outcomes, measure false alerts, omissions, lead time and unavailable support, and require authentic pre-issue report variation to identify reservoir information. Prospective inflow verification requires source-defined full-day means or volumes. Those additions could test transfer; the present study has no measured estimate of avoided agricultural, monetary or livestock loss.

6 Conclusion

Flood Watch achieves raw precision and recall of 80.0% among forty-nine decided September references, with weighted precision 93.8% and recall 66.1%. River-Watch's Bhakra dated-level baseline reduces first-horizon MAE from 0.488 to 0.247 m on 194 paired cases and improves over persistence at all five primary horizons. Pong's small samples and adverse damping comparison restrict transfer. The tested integration adds no correct mapped-positive warnings: all three arms miss ten cases at 0.5, including after re-review. These outcomes support component evidence, while successful integrated future-flood warnings remain unestablished.

Data, code and acknowledgments

Frozen inputs, predictions, reviews, sampling weights and manifests remain in the project. The full companion retains detailed ledgers and secondary plots. Forty reservoir candidate fits and 180 integration scores were independently reconstructed; this shorter paper changes no model, observation, threshold or label.

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